Weixuan Yuan

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Research internship interests: LLM reasoning, post-training, and AI for mathematics and TCS.

Education & Research Positions

ETH Zürich — Ph.D. in Computer Science May 2025–Present
Advisor: Prof. Rasmus Kyng.

ETH Zürich — Research Assistant October 2024–April 2025
Rasmus Kyng's group.

ETH Zürich — M.Sc. in Data Science September 2022–October 2024
Willi Studer Prize (best graduate in the MSc Data Science program, 2025).

Xi'an Jiaotong University — B.Sc. in Mathematics September 2018–July 2022
Honors Youth Program

Selected Research

AuXSpec — Formalizing Programming Problems with LLMs

We study how to train LLMs to turn natural-language programming problems into formal mathematical specifications of correct outputs.

- Proposed the executable-postcondition approach to efficiently test generated specifications; co-designed the Lean specification format.
- Co-developed a multistage LLM-based pipeline for data synthesis, verification, and repair. The team produced 3,037 formalized LeetCode problems and approximately 19.5K SFT samples.
- Contributed to post-training methodology and reward-design discussions and result analysis. The team's 27B model achieved SpecGen pass@4 of 60.3% on AuXSpec-Test and 84.7% on Verina after SFT/RL.

Graph algorithms — FOCS 2025

Co-authored the first deterministic almost-linear-time Gomory-Hu tree algorithm, running in $m^{1+o(1)}$ time. Presented the paper at the IEEE Symposium on Foundations of Computer Science (FOCS 2025). [slides]

Selected Publications

Deterministic Almost-Linear-Time Gomory-Hu Trees
Amir Abboud, Rasmus Kyng, Jason Li, Debmalya Panigrahi, Maximilian Probst Gutenberg, Thatchaphol Saranurak, **Weixuan Yuan**, Wuwei Yuan.
FOCS 2025. [paper]

Optimal subsampling for principal component analysis
Xuehu Zhu[†], **Weixuan Yuan**[†], Zongben Xu, Wenlin Dai.
Science China Mathematics 68, 2993-3016, 2025. [paper]

Optimal subsampling for estimation of dimension reduction directions
Xinru Jia[†], **Weixuan Yuan**[†], Xingqiu Zhao, Xuehu Zhu.
Scandinavian Journal of Statistics 53(1), 575-611, 2026. [paper]

[†] Equal contribution. Author order in theoretical computer science papers is alphabetical. The statistics research was conducted during my undergraduate studies (before 2023).

Under-Review Manuscript

AuXSpec: Learning to Autoformalize Program Specifications via Executable Postconditions
Co-first author; under review at ICLR 2027. Not yet publicly available.

Technical Skills

Programming: Python, Lean 4.
Theory: Algorithms; probability and statistics; optimization, including gradient-based methods.